Nonlinear solutions in Quantum Field Theory

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1. Motivations: Nonlinear classical solutions and their quantization

The first Lagrangian that one encounters in a field theory course is the free scalar field Lagrangian.

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi(x)\partial^\mu\phi(x) - \frac{1}{2}m^2\phi^2(x). \tag{1.1}$$

Where $x \in \mathbb{R}^d$. At the classical level, the EOM are given by the least action principle:

$$(\Box + m^2)\phi(x) = 0. \tag{1.2}$$

This equation has a very particular feature: it is linear! This fact has two major consequences. First, it allows one to write the explicit and complete solution (for free boundary conditions):

$$\phi(x) = \frac{1}{(2\pi)^{d/2}}\int \frac{\mathrm{d}^{d-1}p}{\sqrt{2\omega_p}}\left(a(p)e^{-ip\cdot x} + a^*(p)e^{ip\cdot x}\right). \tag{1.3}$$

Second, the classical solution gives us a suggestion for its quantization. In fact one can impose:

$$a^* \to a^\dagger, \quad [a(p), a^\dagger(p')] = i\delta(p-p'). \tag{1.4}$$

So that the quantum solution for the Heisenberg operator $\phi(x)$ takes the form:

$$\phi(x) = \frac{1}{(2\pi)^{d/2}}\int \frac{\mathrm{d}^{d-1}p}{\sqrt{2\omega_p}}\left(a(p)e^{-ip\cdot x} + a^\dagger(p)e^{ip\cdot x}\right). \tag{1.5}$$

Similarly it is possible to define a free theory for fermions and higher spin theories. Next one would like to define different Lagrangians, in which one does not expect a linear solution. A common example is the "double well Lagrangian":

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi(x)\partial^\mu\phi(x) - \frac{\lambda}{4}\left(\phi^2(x) - v^2\right)^2. \tag{1.6}$$

Where v is a constant. Now there are two alternatives:

1. finding a minimum of the potential and studying the small (quantum) oscillations around it, involving the standard theory of perturbation theory à la Feynman;

2. trying to find classical nonlinear solutions and quantize them in a semiclassical regime.

If the classical solution is nonlinear there is no way to define an exact quantum correspondence trivially, as it was for the free particle. On the other hand it is interesting to ask what kind of intuition these solutions can give us at the quantum level. This is the motivation for the following notes. In the second section we find some classical solutions in one and more field dimensions, and try to give some kind of interpretation. In the third section we focus on 1d fields and see how these solutions can be quantized in the semiclassical approximation. Finally, in the last section we briefly discuss some extensions of the previous results, using the Sine-Gordon model as a special model to observe various phenomenology with analytic and exact solutions.

2. Classical solutions

2.1 One dimension: Kinks

Let us analyse the classical solutions of two particular relativistic 1+1 d models with one field of the form:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi). \tag{2.1}$$

The *double well model* (DW):

$$V(\phi) = \frac{\lambda}{4}\Big(\phi^2(x) - v^2\Big)^2. \tag{2.2}$$

And the *Sine-Gordon model* (SG):

$$V(\phi) = -\frac{m^4}{\lambda}\Big(\cos\Big(\frac{\sqrt{\lambda}}{m}\phi\Big) - 1\Big). \tag{2.3}$$

Even if the latter seems more complicated than the former, the subsequent analysis will show that the integrability property of the SG model allows one to calculate all solutions analytically.

Now we search for stationary non-uniform solutions. Starting with the DW, the problem is equivalent to searching for the functional minima of the energy:

$$U = \int \mathrm{d}x\ \left(\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 + \frac{\lambda}{4}\Big(\phi(x)^2 - v^2\Big)^2\right). \tag{2.4}$$

This gives the ODE:

$$-\frac{d^2\phi}{dx^2} + \lambda(\phi^2 - v^2)\phi = 0. \tag{2.5}$$

Multiplying by $\frac{d\phi}{dx}$ one is able to find the conserved divergenceless current for the steady solution:

$$\frac{d}{dx}\left(-\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 + \frac{\lambda}{4}\Big(\phi(x)^2 - v^2\Big)^2\right) = 0. \tag{2.6}$$

Imposing as asymptotic boundary conditions $\phi(\pm\infty) = \pm v, \quad \partial_x\phi(\pm\infty) = 0$, the previous equation becomes:

$$\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 = \frac{\lambda}{4}\Big(\phi(x)^2 - v^2\Big)^2. \tag{2.7}$$

Which has solution (if $\phi(-\infty) = -v$ and $\phi(+\infty) = v$):

$$\phi_{DW}(x) = v\tanh\left(\frac{m}{\sqrt{2}}(x - x_0)\right). \tag{2.8}$$

In general the previous steps are valid for a generic potential, giving the *virial relation*:

$$\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 = V(\phi). \tag{2.9}$$

Notice that, taking the square root of both sides, the virial relation tells us that it is not possible to have a nontrivial solution whose boundary conditions correspond to the same vacuum. In the case of the SG model, this gives:

$$\phi_{SG}(x) = Nv + \frac{2v}{\pi}\arctan\Big(e^{m(x-x_0)}\Big). \tag{2.10}$$

Where the boundary conditions imposed are $\phi(-\infty) = Nv$ and $\phi(+\infty) = (N+1)v$. This kind of solution in 1d is called *kink*. Switching the boundary conditions, i.e. $v \to -v$, one obtains the *antikink* solution. This classification by boundary conditions, which are indeed not affected by infinitesimal deformations, leads us to reflect on the role of topology in this discussion. In particular it is possible to define a topological current:

$$J^\mu = \frac{1}{2v}\epsilon^{\mu\nu}\partial_\nu\phi. \tag{2.11}$$

Related to the conserved charge:

$$Q = \int \mathrm{d}x \frac{1}{2v}\partial_x\phi = \frac{1}{2v}(\phi(+\infty) - \phi(-\infty)). \tag{2.12}$$

In the case of the kink solution $Q = 1$, for the trivial solution $Q = 0$, and for the antikink $Q = -1$. Notice that this current does not arise from a symmetry of the Lagrangian, as usually happens. Knowing the stationary solution and the Lorentz invariance of the theory, it is possible to obtain a dynamical solution by performing a Lorentz transformation:

$$\phi(x,t) = v\tanh\left(\frac{m}{\sqrt{2}}\gamma(x - x_0 - u(t-t_0))\right). \tag{2.13}$$

Finally let us study the total energy of the solutions. In both cases, the static solution has finite energy. For the DW model one obtains:

$$M_{cl} = \frac{2\sqrt{2}}{3}\frac{m^3}{\lambda}. \tag{2.14}$$

In the dynamical solution, energy transforms like the particle energy $M_u = \gamma M_0$. This suggests the possibility of interpreting these solutions as some type of particle of the theory, which can move in space with velocity u.

2.2 Two and more dimensions: Derrick's theorem

In this section we try to generalize the previous result about finiteness of energy for a classical solution, in the case of more than one field in the lagrangian and for general spatial dimensions for the Minkowski space. For the existence of such solutions, there is a "non-existence theorem" which limits our possibilities considerably.

Teorema 2.1: Derrick's theorem

Given a Lagrangian of the form:

$$\mathcal{L} = \frac{1}{2}G_{ab}(\phi)\partial_\mu\phi^a\partial^\mu\phi^b - V(\phi), \quad a,b = 1,..,d, \quad \mu = 0,...D. \tag{2.15}$$

with G_{ab} a non-negative form and $V(\phi)$ a non-negative potential, then there exist no nontrivial stationary soliton solutions in dimensions $D \geq 3$ with finite energy.

Proof. The energy of the system at a stationary solution is given by the following expression:

$$U(\phi) = \int \mathrm{d}^D x \ \frac{1}{2} G_{ab}(\phi) \partial_i \phi^a \partial^i \phi^b + V(\phi) = I_k + I_V.$$

If $\bar{\phi}(x)$ is a stationary solution, it is a minimum of the energy, and in particular a minimum with respect to the one-parameter family $\bar{\phi}(\lambda x)$. The energy of this family of configurations is given by:

$$\lambda^{2-D} I_k + \lambda^{-D} I_V.$$

Differentiating with respect to λ and evaluating at $\lambda = 1$, the minimum condition leads to the equation:

$$(D-2) I_k + D I_V = 0.$$

If $D = 1$ one obtains the virial relation of the previous section. In $D = 2$ this leads to $I_V = 0$, which is possible only if $V(\phi) = 0$ for all values of ϕ. If $D > 2$ we have a sum of two non-negative quantities equal to zero, which is possible only for a uniform solution at the minimum of the potential. □

One possibility to admit soliton solutions with finite energy consists in adding a gauge field to the Lagrangian:

Teorema 2.2: Derrick's theorem 2

Given a Lagrangian of the form:

$$\mathcal{L} = -\frac{1}{2} \operatorname{tr}(F_{\mu\nu} F^{\mu\nu}) + \frac{1}{2} G_{ab}(\phi) D_\mu \phi^a D^\mu \phi^b - V(\phi), \tag{2.16}$$

with G_{ab} a non-negative form and $V(\phi)$ a non-negative potential, then there exist no nontrivial stationary soliton solutions in dimensions $D \geq 5$ with finite energy.

The proof of this theorem is analogous to the previous case, considering that now the gauge term contributes a term $(D-4) I_G$ to the stationarity condition. Notice that in $D = 4$ only solutions related to the free Yang-Mills theory are allowed.

3. Quantization

3.1 Simple dimensional analysis

Now that we have seen some features of classical solutions we would like to understand whether the classical features hold in the quantum regime. A necessary condition can be obtained by requiring that there exists a characteristic length L, which represents the scale at which we observe the classical solution, such that:

1. L is much smaller than the characteristic length of the classical system, i.e. $L \ll m^{-1}$;
2. in an interval L the fluctuations of the classical field are much larger than the quantum fluctuations.

Regarding the second condition, it is possible to estimate the classical variation as:

$$\Delta\phi \approx \frac{\mathrm{d}\phi}{\mathrm{d}x} L \approx \frac{m^2 L}{\sqrt{\lambda}}. \tag{3.1}$$

For the quantum fluctuation, a rough estimate comes from the two-point correlation function of a free scalar field smeared over a region of width L:

$$\phi_L(x) = \frac{1}{(2\pi L^2)^{n/2}} \int \mathrm{d}^n y \; e^{-(y-x)^2/2L^2} \phi(y). \tag{3.2}$$

$$(\Delta\phi_L)^2_{qu} = \langle 0|\phi_L^2(0)|0\rangle = \int \frac{\mathrm{d}^n k}{(2\pi)^n} \frac{e^{-k^2L^2}}{2\sqrt{k^2+m^2}}. \tag{3.3}$$

For $n = 1$ one gets:

$$(\Delta\phi_L)_{qu} \sim \sqrt{\log(1/mL)}. \tag{3.4}$$

Comparing the two expressions:

$$\sqrt{\log(1/mL)} \ll \frac{m^2 L}{\sqrt{\lambda}}, \tag{3.5}$$

which, combined with $L \ll m^{-1}$, is possible only if λ/m^2 is small. This calculation loses a solid quantum interpretation in the strong coupling regime, and confines us to work in the weak coupling regime.

3.2 Zero modes and collective coordinates

In this section we gain some intuition about how to quantize the static kink solution in 1+1 d in a semiclassical approximation, obtaining the corresponding energy. The main idea is to separate the field and the Lagrangian into different parts:

$$\phi(x,t) = \phi_{cl}(x) + \eta(x,t), \quad L = L_{cl} + L_{quad} + L_{int}. \tag{3.6}$$

The field is divided into the classical solution and the quantum fluctuation, whereas the Lagrangian is divided into the classical part, the quadratic term in the fluctuations, and the higher-order terms. The approximation consists in neglecting higher-order terms and working only with the quadratic Lagrangian:

$$L_{quad} = \int \mathrm{d}x \left(\frac{1}{2}(\partial_\mu \eta)^2 - \frac{1}{2} V''(\phi_{cl}) \eta^2 \right). \tag{3.7}$$

We now find the normal modes via:

$$\eta(x,t) = f_j(x) e^{-i\omega_j t} \implies \left(-\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V''(\phi) \right) f_j(x) = \omega_j^2 f_j(x). \tag{3.8}$$

The energy of the quadratic term is now given by the formula for the harmonic oscillator:

$$E_{quad} = \sum_j (n_j + 1/2)\omega_j. \tag{3.9}$$

In order to concretely obtain the first excitation of a quantum soliton solution there are a series of technical steps, which are briefly described below.

- Add to the Lagrangian a constant term to give a finite value to the vacuum, and a counterterm for the renormalization scheme;
- discretize the number of modes N and impose periodic boundary conditions. At the end of the calculation, take $N \to \infty$;
- calculate the modes associated to bound states and scattering, corresponding to the kink and the vacuum solutions, and then subtract them, imposing $n_j = 0$ for both (adding nonzero n_j can be interpreted as adding elementary particles to the kink).

The final result for the DW model is:

$$M_{kink} = \frac{2\sqrt{2}}{3}\frac{m^3}{\lambda} + \left(\frac{\sqrt{6}}{12} - \frac{3\sqrt{2}}{2\pi}\right) m + \ O\left(\frac{\lambda}{m}\right). \tag{3.10}$$

Where the first term comes from the classical solution whereas the second comes from the two bound state modes. Notice that for a general kink solution there always exists a *zero mode*. This can be easily seen by noticing that in equation (3.8), with $\phi = \phi_{cl}(x)$, the function $\frac{\mathrm{d}\phi_{cl}(x)}{\mathrm{d}x}$ is always a solution:

$$\left(-\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V''(\phi_{cl})\right)\frac{\mathrm{d}\phi_{cl}(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{\mathrm{d}^2\phi_{cl}}{\mathrm{d}x^2} + V'(\phi_{cl})\right) = 0. \tag{3.11}$$

For this kind of solution an interpretation in terms of excitations may be problematic, because every multiparticle state contributes zero energy. It is possible to give an explanation of this phenomenon by observing that:

$$\frac{\mathrm{d}\phi_{cl}}{\mathrm{d}x} = -\frac{\mathrm{d}\phi_{cl}}{\mathrm{d}x_0}. \tag{3.12}$$

In other words, the zero mode is related to the symmetry of spatial translation which holds for the Lagrangian and is broken by the soliton. Notice the analogy with the massless Goldstone bosons, produced via symmetry breaking phenomena. It is possible to show that, if the field is decomposed as:

$$\phi(x,t) = \phi_{cl}(x - z(t)) + \eta(x - z(t)), \tag{3.13}$$

the quadratic Lagrangian can be written as:

$$L_{quad} = \frac{1}{2}M_{cl}\dot{z}^2 + \frac{1}{2}\sum_j \left(\frac{1}{2}\dot{c_j}^2 - \frac{1}{2}\omega_j^2 c_j^2\right). \tag{3.14}$$

This form introduces as the leading term of the energy a kinetic energy associated to the kink itself.

$$E = M_{kink} + \frac{P^2}{2M_{cl}} + O(P^4) + O(\lambda). \tag{3.15}$$

A more refined study can show that the first two classical kinetic terms are the leading order of the expansion of the relativistic expression $\sqrt{M_{kink}^2 + P^2}$.

3.3 Possible further questions

Other possible themes and questions can be raised after setting up these arguments: the non-linear classical solutions and their quantization. For brevity, below we list some of them, as a

reflection on various possibilities.

- What about kinks solutions in spacetime with more than 1 spatial dimension? What symmetries do solitons preserve and which do they break?
- What about the introduction of fermions in the theory? Are there any zero modes in their quantization?
- What about gauge symmetry and solitons?

4. Some analytical developments: the Sine-Gordon Model

Now we discuss the other possible classical and quantum solutions obtained by combining solitons and antisolitons. In general we can have two kinds of solutions: scattering-like solutions and bound states, depending on how they behave asymptotically in space and time. For scattering solutions of two solitons, we expect that at $t \to -\infty$ we see two well-separated solitons, at finite t there is a mixed solution, and at $t \to +\infty$ we obtain again two separated solitons. In general there is no exact symmetry between the $\pm\infty$ solutions, because after the scattering part of the energy is converted into radiation, i.e. oscillations of the field (like electromagnetic waves in electromagnetism). However, for particular models this does not happen, and for this reason the analysis of solutions is much simpler. In 1+1d an example of such models is the Sine-Gordon model, defined in the second section. In the following subsection we examine the explicit solutions.

4.1 Multikink solutions

The SG theory has two non-topologically trivial fundamental solutions, the kink and the antikink. The scattering of two kink states can be written as:

$$\phi(x,t) = \frac{2v}{\pi}\tan^{-1}\left(u\sinh\left(\frac{mx}{\sqrt{1-u^2}}\right)\cosh^{-1}\left(\frac{mut}{\sqrt{1-u^2}}\right)\right). \tag{4.1}$$

The energy is:

$$E = \frac{16m^3}{\lambda\sqrt{1-u^2}} = \frac{2M_{cl}}{\sqrt{1-u^2}}. \tag{4.2}$$

Analyzing the type of interactions between the kinks one can show that they feel repulsive interactions, which generate a net time advance in the translation of $\Delta t = -\frac{2\sqrt{1-u^2}}{mu}\ln(|u|)$. In the case of the kink-antikink solution, we obtain:

$$\phi(x,t) = \frac{2v}{\pi}\tan^{-1}\left(\frac{1}{u}\sinh\left(\frac{mut}{\sqrt{1-u^2}}\right)\cosh^{-1}\left(\frac{mx}{\sqrt{1-u^2}}\right)\right). \tag{4.3}$$

In this case one can show that the interaction between kink and antikink is attractive. This suggests that a bound state solution may exist. In fact, writing $u = is$ one obtains:

$$\phi(x,t) = \frac{2v}{\pi}\tan^{-1}\left(\frac{1}{s}\sin\left(\frac{mst}{\sqrt{1+s^2}}\right)\cosh^{-1}\left(\frac{mx}{\sqrt{1+s^2}}\right)\right). \tag{4.4}$$

This kind of solution is called *breather*, and the corresponding energy is:

$$E_{breather} = \frac{2M_{cl}}{\sqrt{1+s^2}} < 2M_{cl}. \tag{4.5}$$

Passing to the quantized spectrum, an analysis similar to that performed before, considering the stationary phase plus second-order approximation, gives the following masses:

$$M_n = \frac{16m}{\gamma'} \sin\left(\frac{n\gamma'}{16}\right), \quad n = 1, 2, ..., < \frac{8\pi}{\gamma'}. \tag{4.6}$$

Where $\gamma' = \frac{\lambda}{m^2}\left(1 - \frac{\lambda}{m^2}\right)$. The mass of the kink is also corrected by the factor γ':

$$M_{kink} = \frac{8m}{\gamma'}. \tag{4.7}$$

Even though this was a first-order calculation, the same result holds at each perturbative level. Note that in the $\lambda \to 0$ limit the masses become degenerate with the elementary bosons. A more detailed analysis can show that they are in fact the corresponding bound states of bosons.

4.2 Strong coupling regime

At the start of the quantization section we stated that the classical solution can be a good approximation to its quantum counterpart only in the weak coupling regime. However, due to the property of the SG model of being an integrable model, the previous result holds for any value of the coupling, even if the connection with the classical theory is no longer clear. To further investigate the model we now show a relation with a *dual* model, the *Thirring model*, given by the 1+1d Lagrangian:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - M)\psi - \frac{g}{2}(\bar{\psi}\gamma^\mu\psi)^2. \tag{4.8}$$

The breather spectrum can be rewritten from (4.6) as:

$$M_n = 2M_{kink} \sin\left(\frac{n\pi}{2} \frac{\lambda/(8\pi m^2)}{1 - \lambda/(8\pi m^2)}\right), \quad n < \frac{8\pi m^2}{\lambda} - 1. \tag{4.9}$$

For $\lambda > 2\pi m^2$ the kink is lighter than the breather, whereas for $\lambda > 4\pi m^2$ there are no more breather solutions. Just below this threshold, defining δ by:

$$\frac{\lambda}{4\pi m^2} = \frac{1}{1 + \delta/m}. \tag{4.10}$$

The mass of the breather reads:

$$M_1 = M_{kink}\left(2 - \delta^2 + \frac{4\delta^3}{\pi} + O(\delta^4)\right). \tag{4.11}$$

Curiously, the mass of the fermion-antifermion bound state in the Thirring model in the weak coupling limit is:

$$M_{bound} = M\left(2 - g^2 + \frac{4g^3}{\pi} + O(g^4)\right). \tag{4.12}$$

Which is exactly the same expression obtained in the previous threshold limit. One is thus led to conjecture the following correspondence:

$$\frac{\lambda}{4\pi m^2} = \frac{1}{1+g/m}. \tag{4.13}$$

So that the $\lambda/m^2 \to 0$ weak coupling limit for the SG model corresponds to the $g \to \infty$ limit for the massive Thirring model. It was shown that this correspondence is actually true for all values of λ and g, and there is a full reformulation of the two theories:

$$\text{kink} \iff \text{elementary } \psi \tag{4.14}$$

$$\text{antikink} \iff \text{elementary } \bar{\psi} \tag{4.15}$$

$$\text{elementary } \phi \iff \text{bound state } \bar{\psi}\text{-}\psi \tag{4.16}$$

$$\text{topological charge} \iff \text{fermion number.} \tag{4.17}$$

This is interesting because we have found two different theories indistinguishable from an experimental point of view. This leads to two final comments:

- We have two equivalent theories in which the soliton solutions are equivalent to particle states, suggesting that there should not really be a conceptual difference between the two;
- the perturbative weak coupling regime of one theory gives information about the strong coupling regime of the other, and vice versa.

5. References

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